

NOTES — Point of Sale (POS) system:

Tech Stack — React, Python, Postgres

Phase 1: Requirements restated

A simple ordering and payment system to digitalise the system.

An AI chatbot has been included to explain the menu to the users.

Ambiguities found + assumptions made

- **No auth/roles defined.** Assumption: No authentication explicated stated.
- **Persistence requirement stated.** Persistence was required however due to time constraints, the code for logging the transaction have been commented.
- **No UI fidelity specified.** Assumption: function over polish — clean and usable, not pixel-perfect.

Architecture sketch

- `/chat - { message: str }`
- `/payment - : { success: bool transactionId: str error: str | None }`

What I'm cutting and why

Reasoning: the core triage loop (add → payment → response) for payment.

Product prioritization

Key decisions

- **DB hardcoded however commented** - Due to complexity and time window
- **Chatbot reply hardcoded with same response due API key error, limitation**
- **No error handling**

AI prompt log

1. Write a simple code for restaurant menu and add items to carts in react ts
2. Create a boilerplate FastAPI server
3. Create Readme for frontend and backend
4. Python backed for ai chat

- Main user interface

Hi! Ask me about the menu or what's new today.

Menu

Margherita pizza — \$12.50

Caesar salad — \$8.00

Grilled salmon — \$18.00

Cart

Total: \$0.00

AI-agent workflow screenshots,

Hi! Ask me about the menu or what's new today.

Hello

assistant

Hello

Ask...



- Key user flows

Cart

Margherita pizza x1

Caesar salad x1

Total: \$20.50



Cart

Total: \$0.00

Payment successful!